ACTIVIDAD 1

INTEGRANTES:

CABARCAS FREDYS

MUÑOZ BRIYIT

BENAVIDES SAMUEL

BENITEZ IVAN

BONEU JOSUAT

DOCENTE:

VASQUEZ ALBERTO

GRUPO:

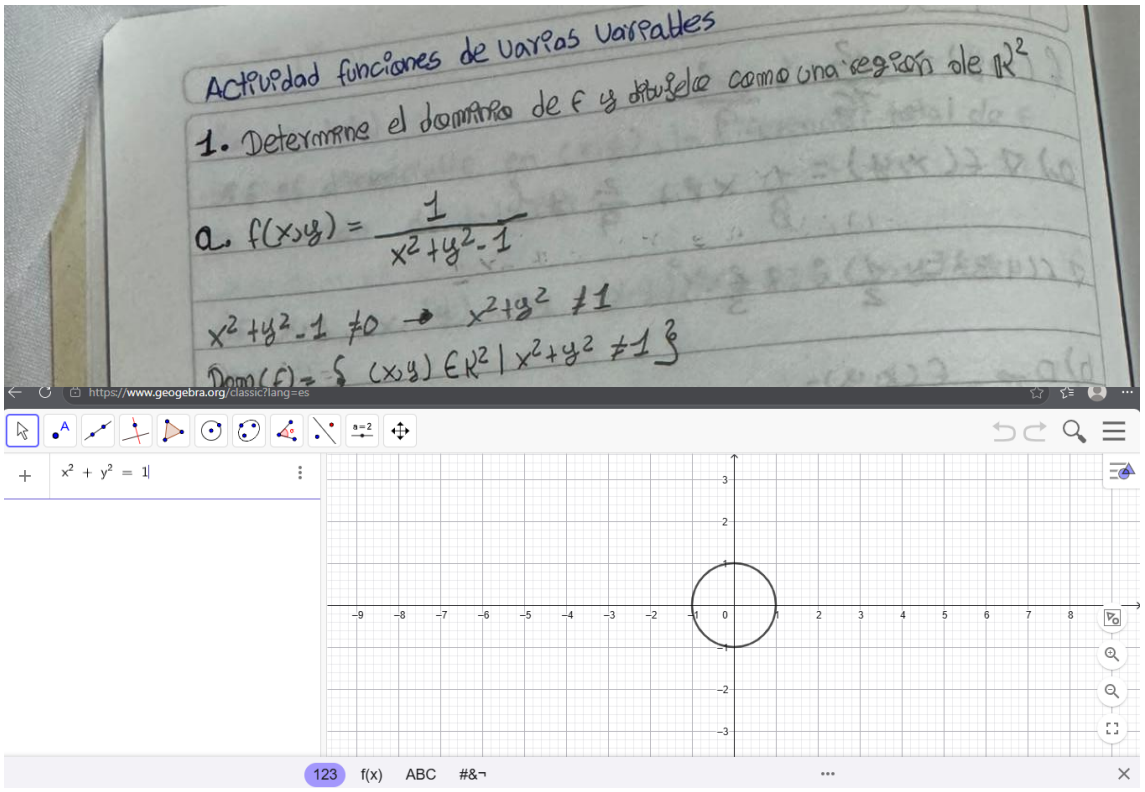
7_GSIN_GRTC_G2

PROGRAMA:

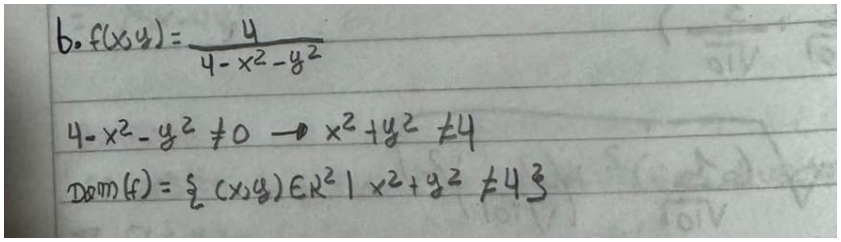
TECNOLOGÍA EN GESTIÓN DE SISTEMAS INFORMÁTICOS

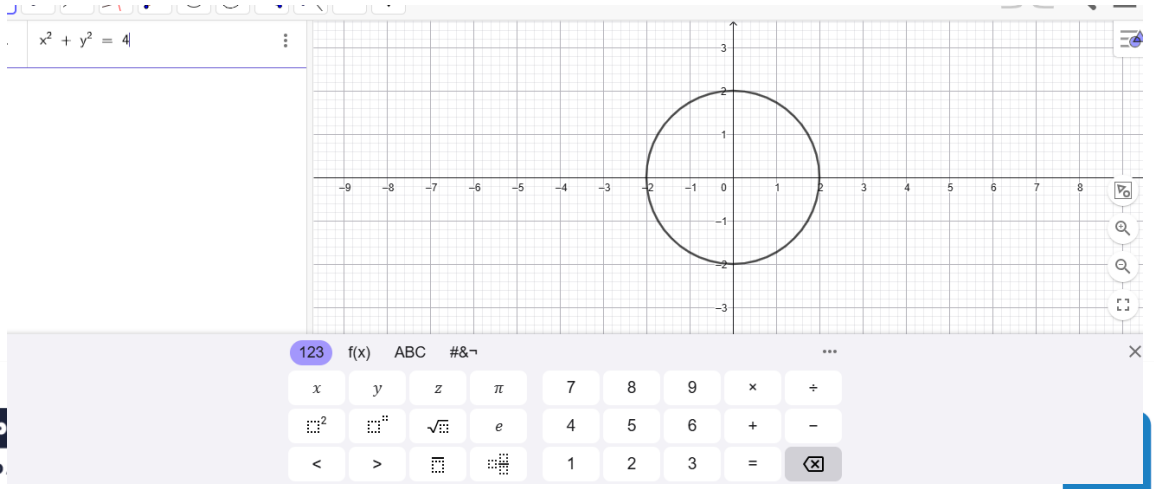
1. Determine el dominio de f y dibujelo como una region de R

A.

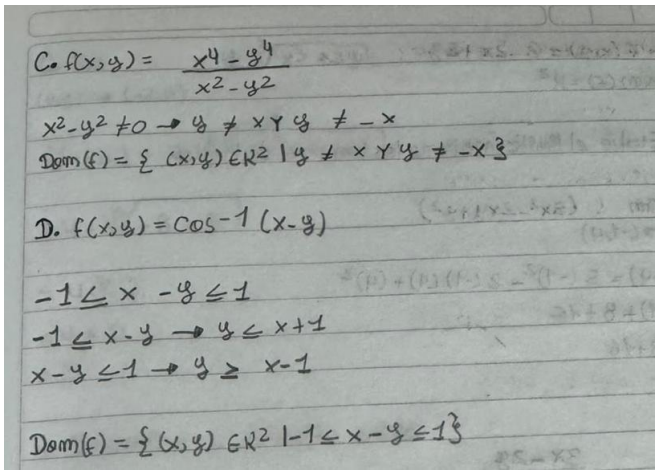


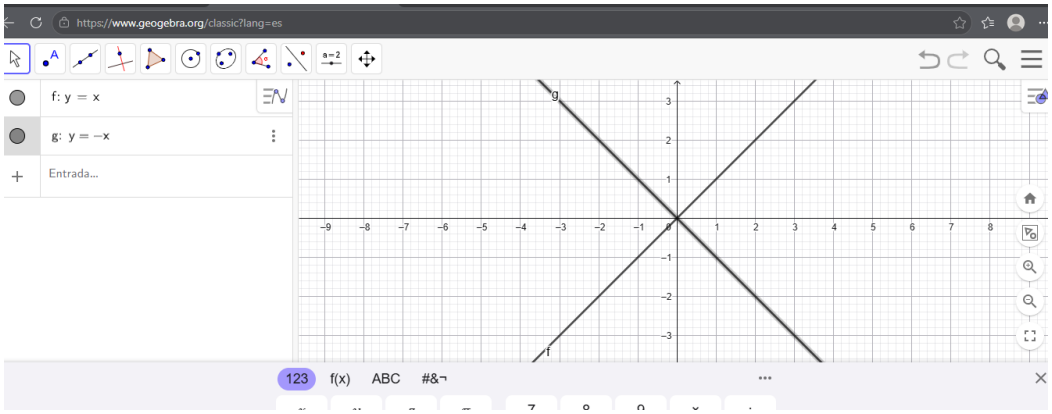
B.



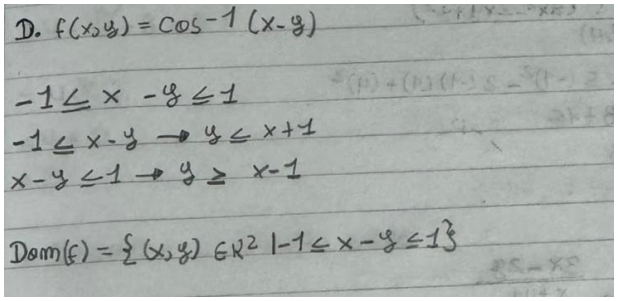


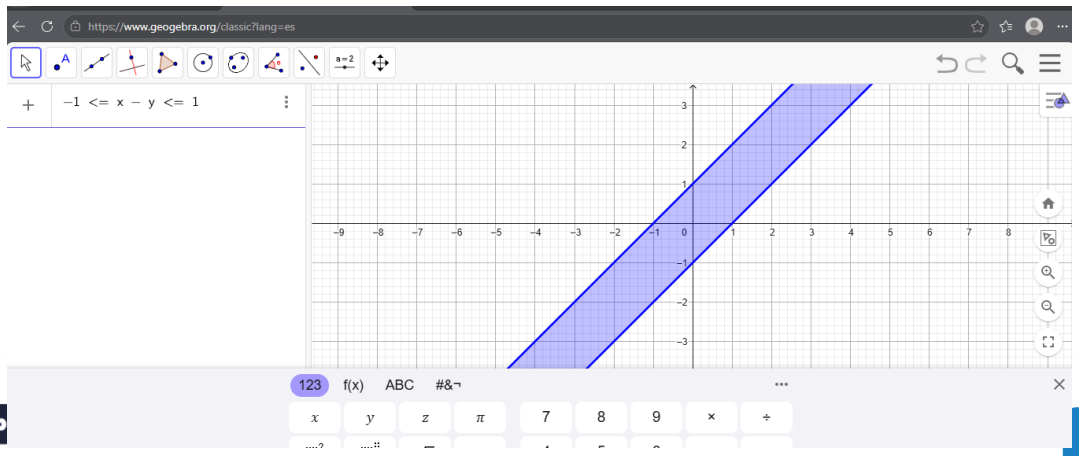
C.


C. $f(x,y) = \frac{x^4 - y^4}{x^2 - y^2}$
$x^2 - y^2 \neq 0 \rightarrow y \neq x \; y \neq -x$
$Dom(f) = \{(x,y) \in R^2 \mid y \neq x \; y \neq -x\}$
D. $f(x,y) = \cos^{-1}(x-y)$
$-1 \leq x - y \leq 1$
$-1 \leq x - y \rightarrow y \leq x+1$
$x - y \leq 1 \rightarrow y \geq x-1$
$Dom(f) = \{(x,y) \in R^2 \mid -1 \leq x - y \leq 1\}$


https://www.geogebra.org/classic?lang=es
f: y = x
g: y = −x
Entrada...
123
f(x)
ABC
#&¬

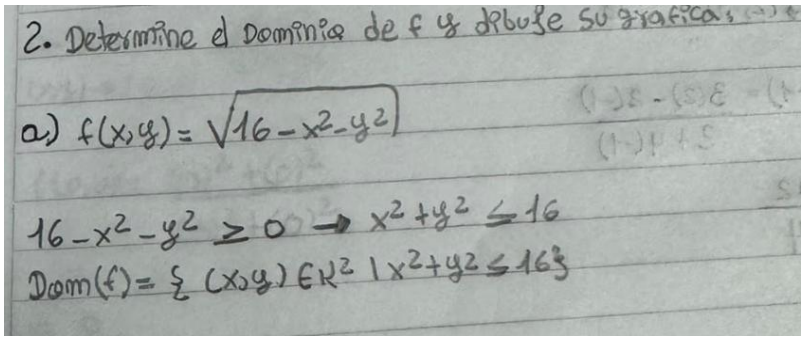
D.

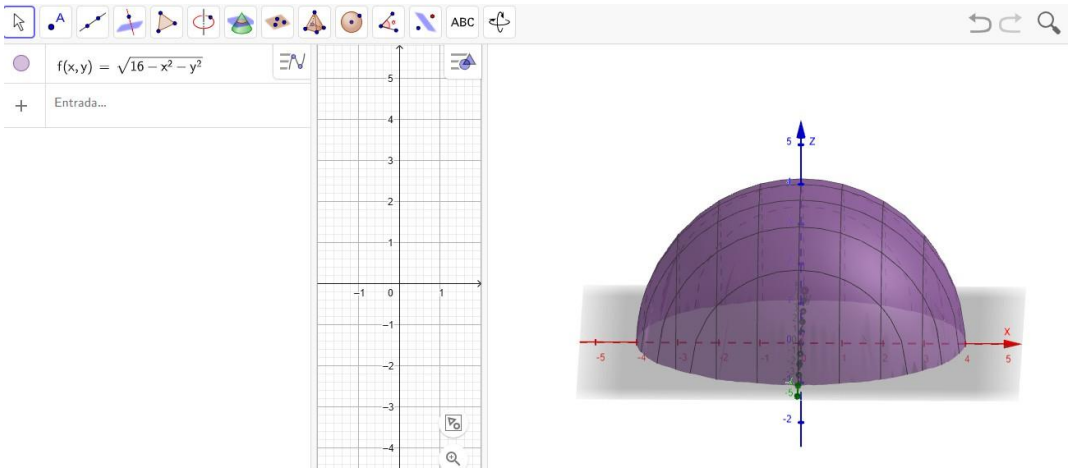

D. $f(x,y) = \cos^{-1}(x-y)$
$-1 \leq x - y \leq 1$
$-1 \leq x - y \rightarrow y \leq x+1$
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$Dom(f) = \{(x,y) \in R^2 \mid -1 \leq x - y \leq 1\}$


https://www.geogebra.org/classic?lang=es
−1 <= x − y <= 1
123
f(x)
ABC
#&¬

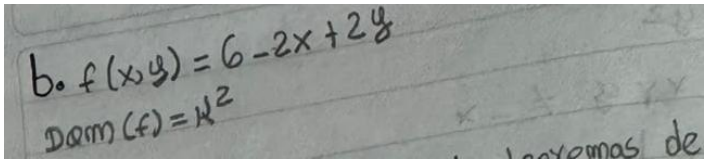
2. Determine el dominio de f y dibuje su gráfica:

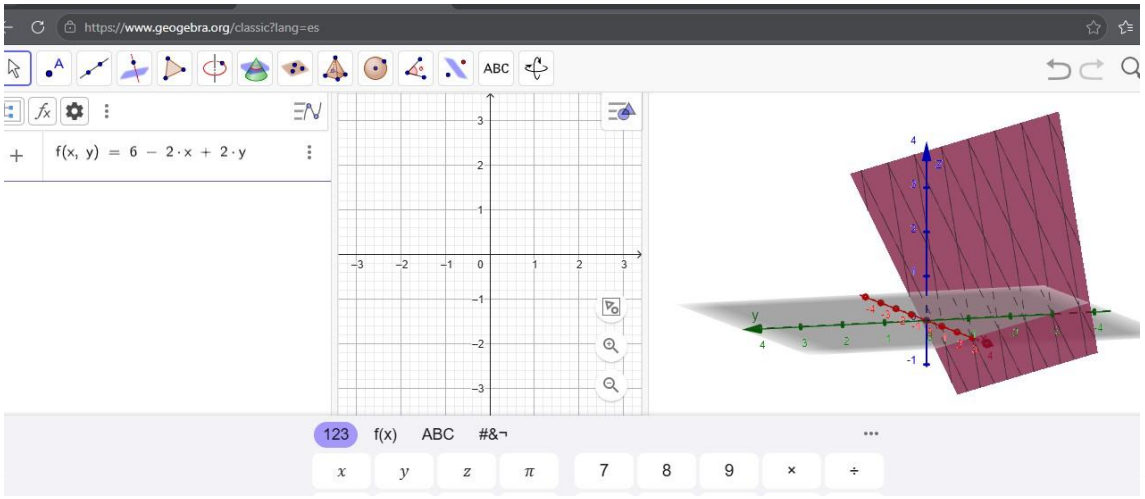
A.



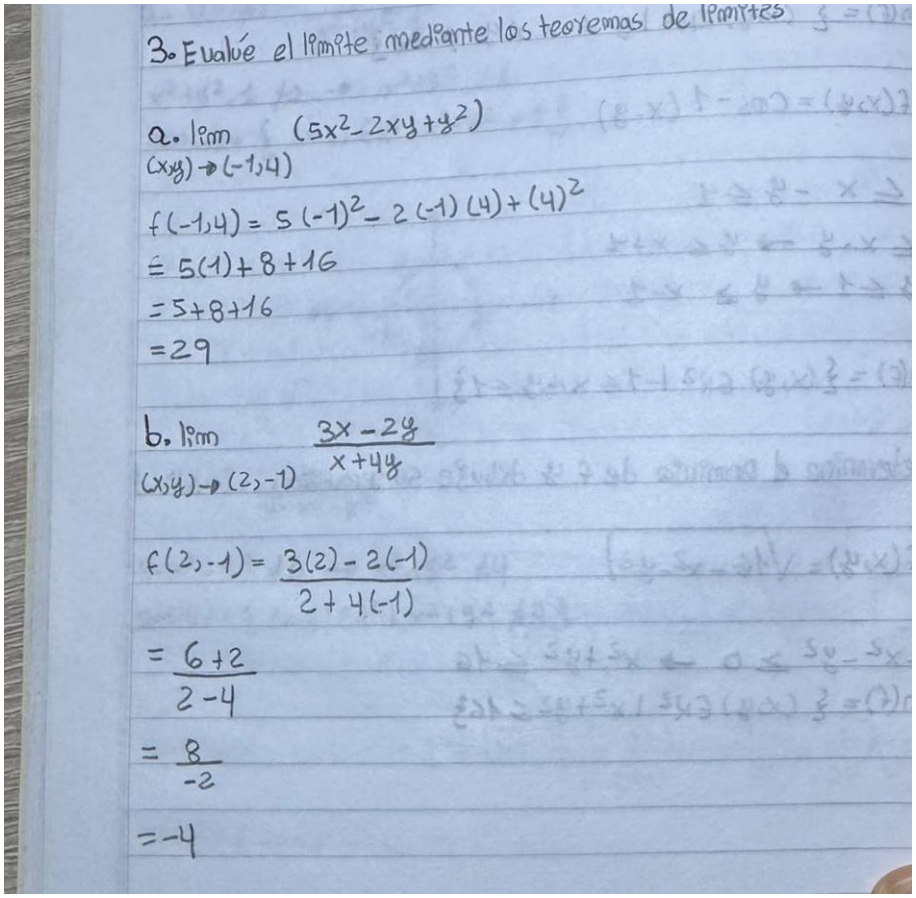


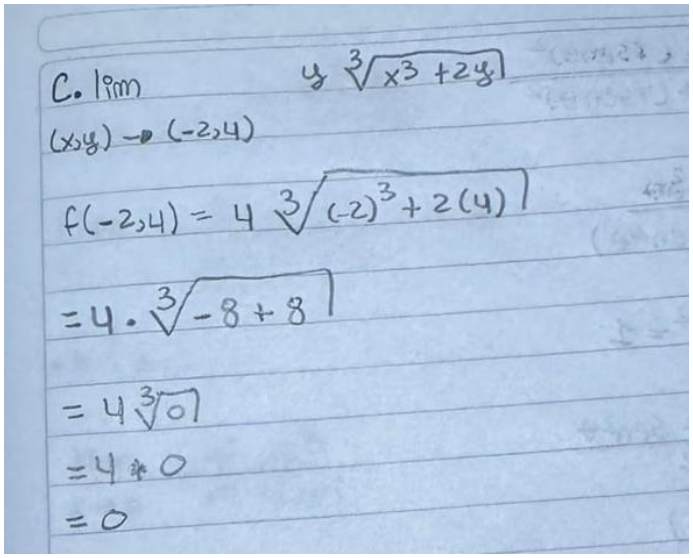
B.





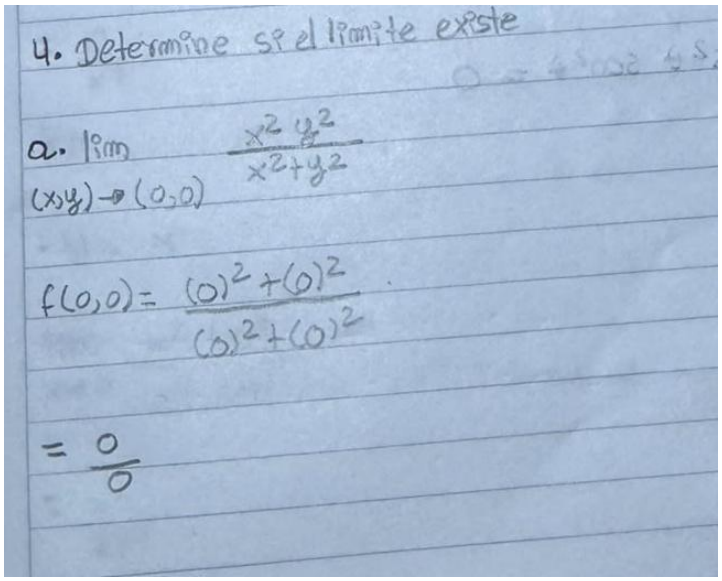
3. Evalué el límite mediante los teoremas de límites:

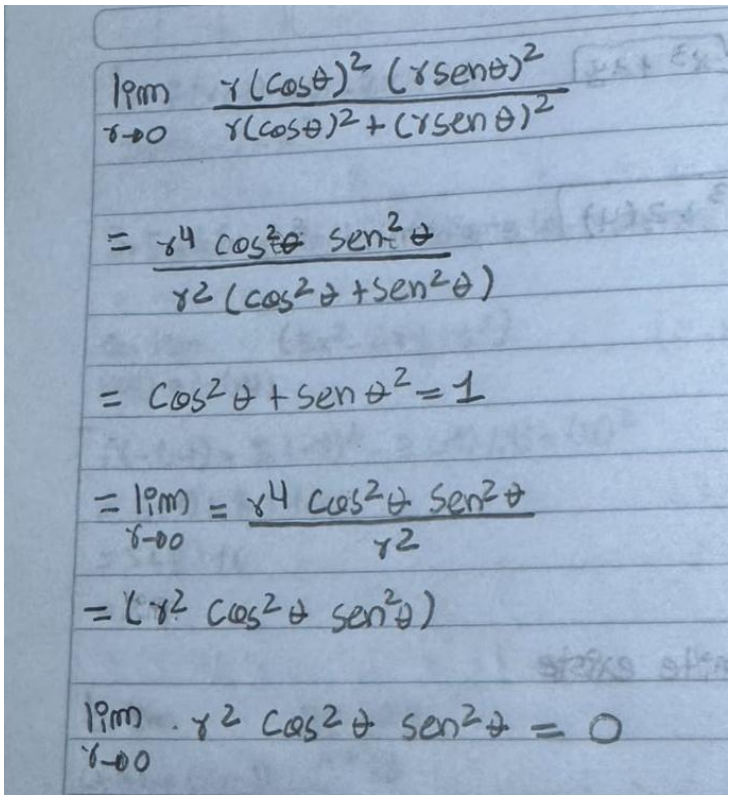




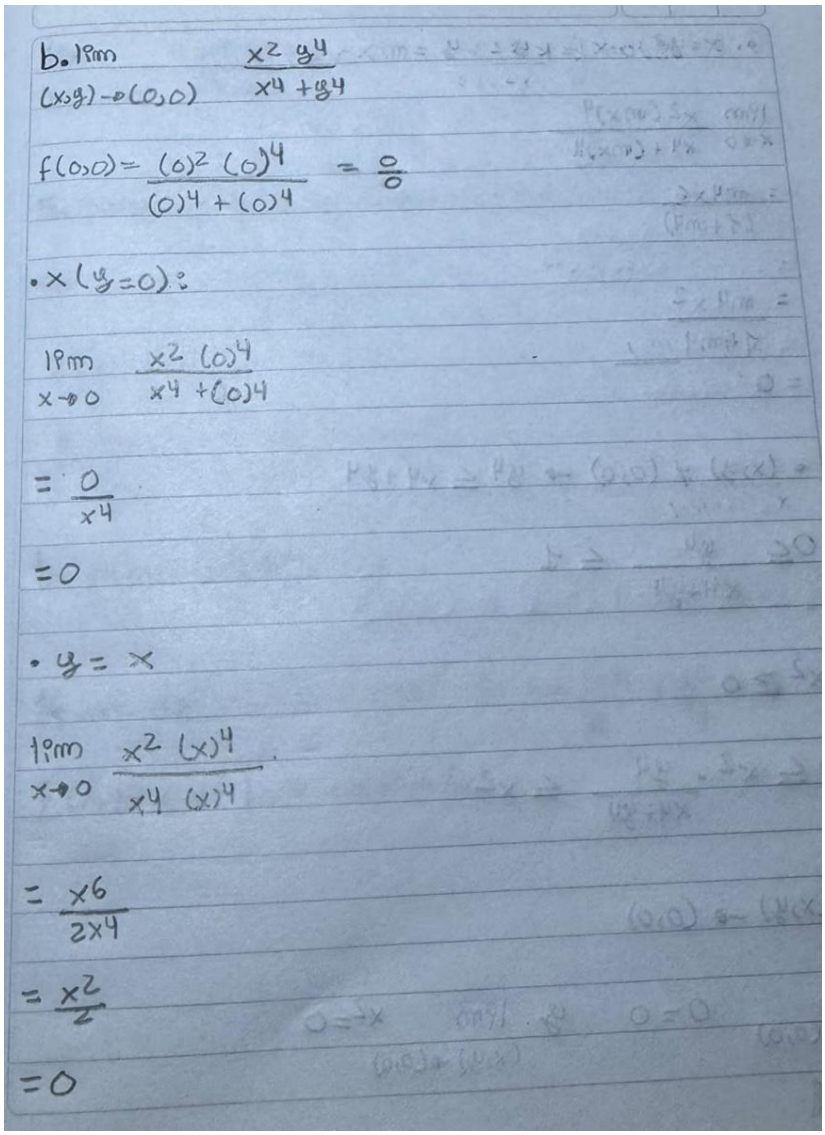
4. Determine si el límite existe:

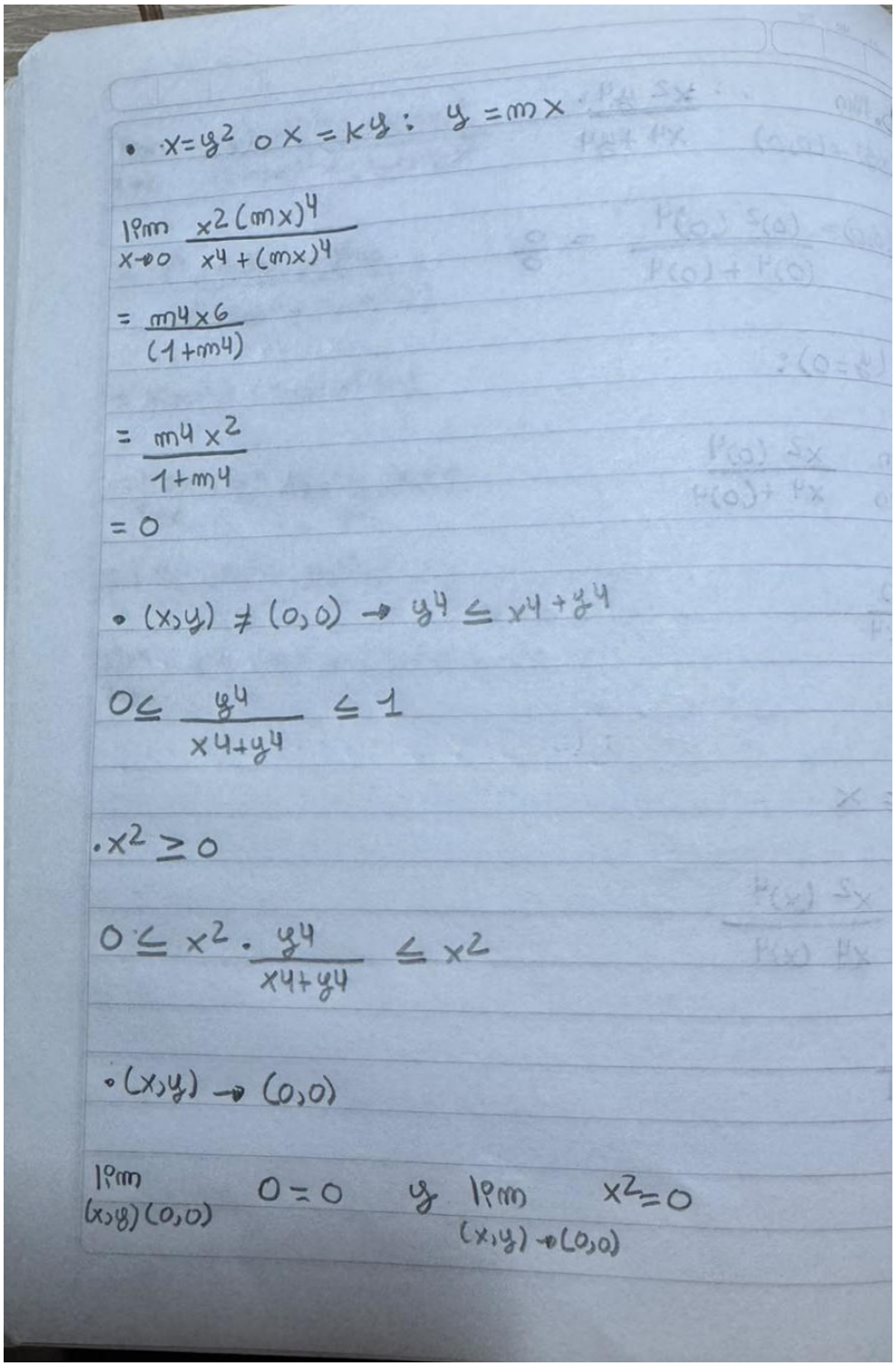
A.

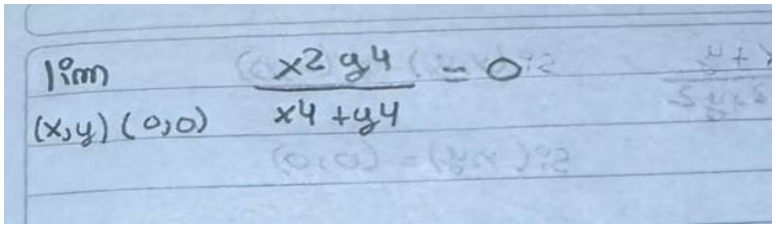




B.


b. lim (x,y)→(0,0) x²y⁴ / (x⁴+y⁴)
f(0,0) = (0)²(0)⁴ / (0)⁴+(0)⁴ = 0/0
• x (y=0):
lim x→0 x²(0)⁴ / x⁴+(0)⁴
= 0/x⁴
=0
• y = x
lim x→0 x²(x)⁴ / x⁴(x)⁴
= x⁶/2x⁴
= x²/2
=0


• x = y² o x = ky : y = mx
lim x→0 x²(mx)⁴ / x⁴ + (mx)⁴
= m⁴x⁶ / (1+m⁴)
= m⁴x² / 1+m⁴
= 0
• (x,y) ≠ (0,0) → y⁴ ≤ x⁴ + y⁴
0 ≤ y⁴ / x⁴+y⁴ ≤ 1
• x² ≥ 0
0 ≤ x² · y⁴ / x⁴+y⁴ ≤ x²
• (x,y) → (0,0)
lim (x,y)(0,0) 0 = 0 y lim (x,y)→(0,0) x² = 0


lim (x,y)(0,0) x²y⁴ / x⁴+y⁴ = 0

5. Determine todos los puntos en los que la función es continua:

